Data Driven Control Integration Week

Group 8 (MPC)

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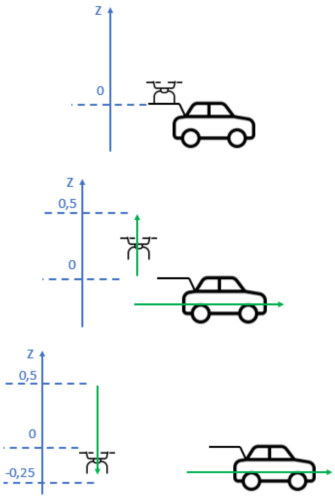
Project Objectives

Available Resources:

- Dataset of vertical acceleration.
- MATLAB/Simulink drone simulator.

Objectives:

- Validate the dataset.
- Identify the model.
- Develop a control law.



Dataset Validation

Cleaning the data

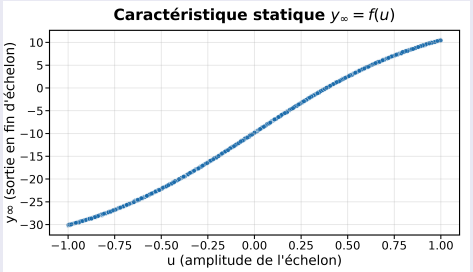
Remove lines that includes:

- Frozen sensor for 2+ seconds
- 5+ diverging values

Normalisation

- Directly in the training

Static characteristic



$y_\infty = f(u)$ after eliminating corrupted runs.

System Modeling

Linear Model

State Function

First-order lag driven by $N(u)$:

$$\begin{cases} \dot{x}(t) = -\dfrac{x(t)}{\tau} + \dfrac{N(u(t))}{\tau} \\ a_z(t) = x(t) \end{cases}$$

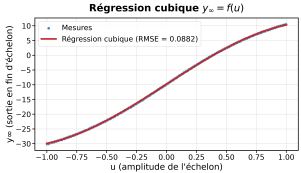
Parameter Identification

- **Static characteristic** $N(u)$: 3rd-order polynomial identified on steady-state regimes:

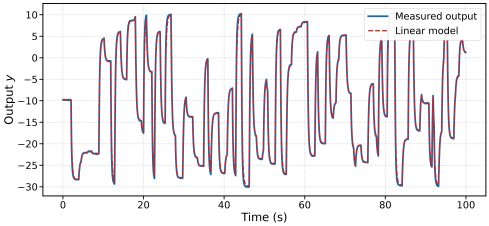
$$a_z = N(u) = a_3 u^3 + a_2 u^2 + a_1 u + a_0$$

- **Tau identification**

$a_3 \approx -6.05$, $a_2 \approx 1.1\times10^{-3}$, $a_1 \approx 26.1$, $a_0 \approx -9.81$, $\tau \approx 0.22$ s



Linear Model Validation



First RNN

Sequence-to-sequence architecture

Parameters

- One scalar input u_t and one scalar output $\hat{y}_t$
- Hidden state dimension: 16
- Early stopping to avoid overfitting

State representation

$$h_t = \tanh(W_{ih}u_t + W_{hh}h_{t-1} + b)$$

$$\hat{y}_t = W_{hy}h_t + b_y$$

First RNN

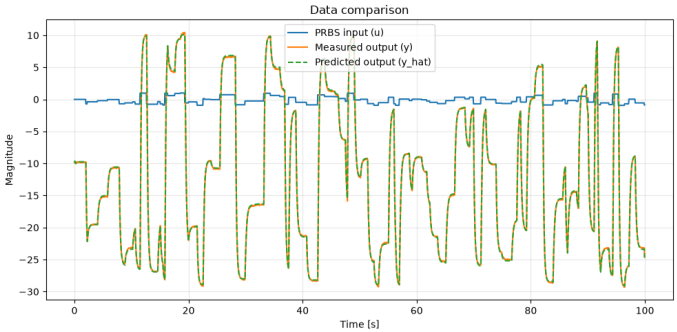
Results

Numerical results

- Validation loss: 3.93×10^{-4}
- Test loss: 4.01×10^{-4}

Training criterion

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$



Z-axis control

Designing the reference trajectory

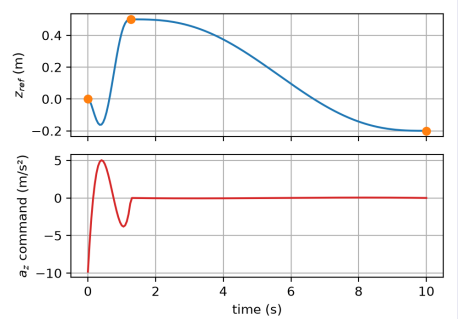
Principle

- **Input**: waypoints z_i and passage times t_i
- One **quintic polynomial** per segment $[t_i, t_{i+1}]$, imposing z, $\dot{z}$ and $\ddot{z}$ at both ends (jerk free)
- Null speed at waypoints, null acceleration on arrival $\Rightarrow$ $\ddot{z}$ continuous
- **Output**: $z_{ref}(t)$ and the acceleration command $\ddot{z}_{ref}(t)$

Results

Smooth analytic reference; overshoot beyond waypoints removed by shortening segments

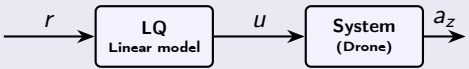
Example: $z = 0, 0.5, -0.2$ m at $t = 0, 5, 10$ s



LQ control

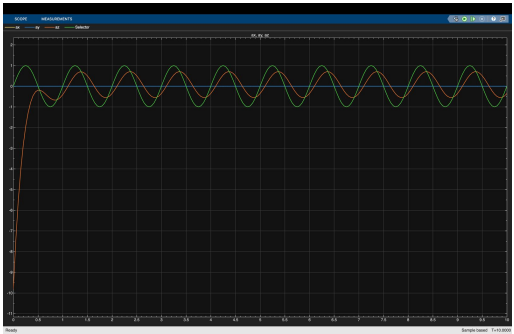
Linear-quadratic control design

Control principle



Performance

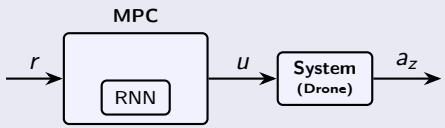
- Global trajectory following
- Wrong amplitude + phase shift
- Missing prediction



MPC control

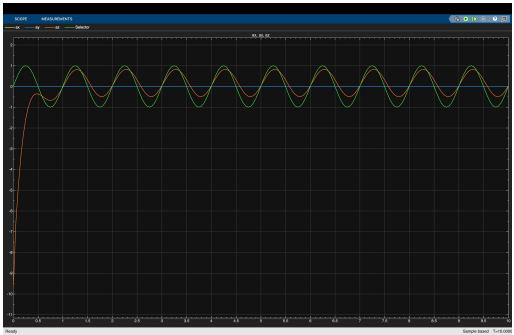
Model predictive control design

Control principle



Performance

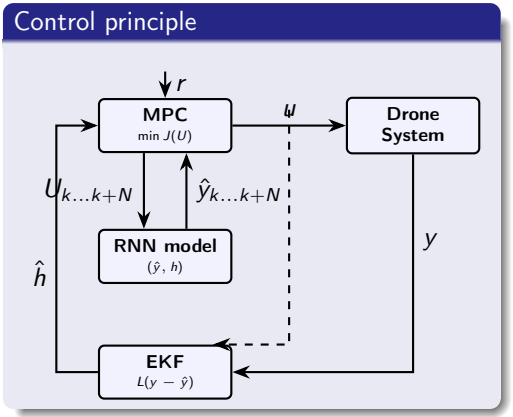
- Good overall tracking.
- No phase shifting good prediction.
- Full amplitude not reached.
- Only internal state.

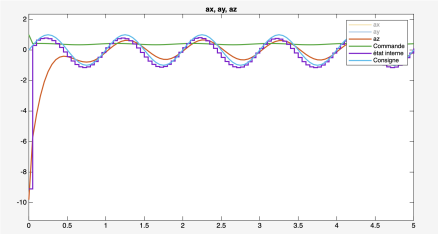


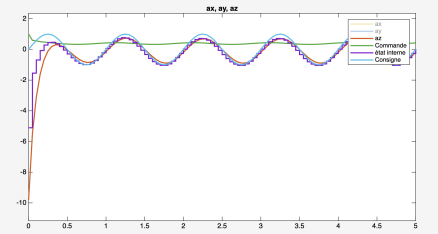
MPC with EKF

State estimation and predictive control

Control principle

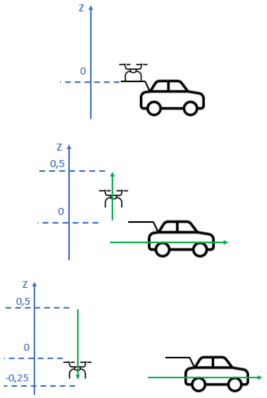


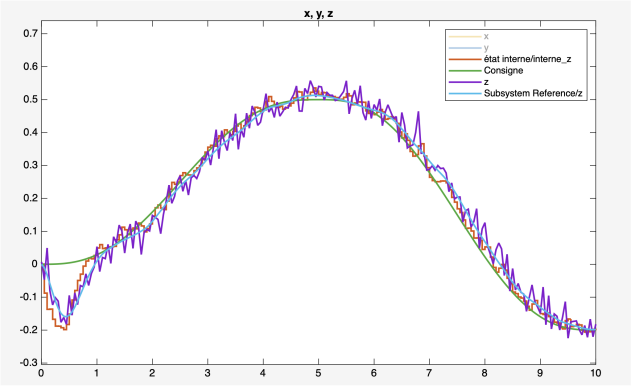




Requested command

MPC control with EKF feedback





Comparison

LQ, MPC and MPC with EKF

LQ control with Linear Model

- Simple and very fast.
- Lack prediction.
- Phase shifting.

MPC control with RNN

- Prediction.
- Internal state diverges from reality.
- No feedback loop.

MPC with EKF

- Feedback from measurements.
- Position control.

Data generation

PRBS excitation in the 3D drone workspace

Objective

Generate sufficiently rich input/output trajectories to identify a nonlinear dynamic model of the drone.

- Excite the four actuator commands around hover
- Capture the coupled 3D dynamics (12 variables)
- Store synchronized inputs and measured outputs

Simulation setup

- Sampling time: $T_s = 0.05\,\mathrm{s}$

APRBS modes

We don't directly generate u inputs (unusable data).
Four independently generated piecewise-constant modes:

$$\delta = [\delta_T, \delta_r, \delta_p, \delta_y]$$

Each level is sampled uniformly and held for a random duration between 1 and 3 seconds.

Initial phase

The excitation starts after $t_{\mathrm{start}} = 2\,\mathrm{s}$, allowing the drone to stabilize at hover.

Data generation

From modal excitation to motor commands

Mixer matrix

The four modal perturbations are mixed into the actuator commands:

$$U = U_{\text{hover}} + \delta \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}^T$$

The resulting commands are clipped to the allowed range.

Input/output data

- Inputs: $U_1, \ldots, U_4$
- Outputs: angles, angular velocities, accelerations and angular accelerations
- One trajectory is stored as a sequence of N time steps
- Different random seeds improve excitation diversity

Appendix: System Modeling

Linear Model: Least Square Method

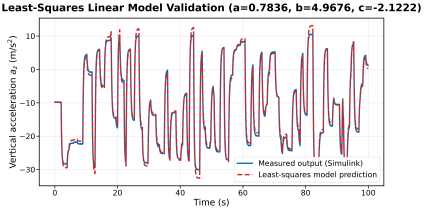
Model equations

$$y_k(t+1) = a\,y_k(t) + b\,u_k(t) + c$$
$$= \varphi_k^{\mathsf{T}}\theta.$$

$$\varphi_k^{\mathsf{T}} = \begin{bmatrix} y_k(t) & u_k(t) & 1 \end{bmatrix} \qquad \theta = \begin{bmatrix} a & b & c \end{bmatrix}^{\mathsf{T}}.$$

The least-squares estimate is

$$\theta^{\star} = \left(\Phi^{\mathsf{T}}\Phi\right)^{-1}\Phi^{\mathsf{T}}Y.$$



Appendix: System Modeling

Linear Model with Constant Steady-State Gain

State Function

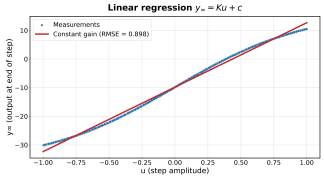
First-order lag driven by a constant gain K and offset c: $\dot{v}(t) = \frac{1}{\tau}\big(Ku(t) - v(t)\big)$, i.e. $\tau\,\dot{v} + v = Ku + c$.

Parameter Identification

- **Static gain** K: affine least-squares fit on steady-state regimes:

$$v(t) = K\,u(t) + c, \qquad K \approx 22.5,\ c \approx -9.8$$

- **Tau identification**



Linear Model Validation

